Analytical Investigation of Nonlinear Distribution Mechanisms in Fluid Dynamics through the Boiti–Leon–Manna–Pempinelli Equation

Article | Published: 26 September 2025
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Abstract

The 3 + 1-dimensional Boiti-Leon-Manna-Pempinelli equation, a well-known higher-dimensional extension of the shallow-water wave model, has been taken into account. This article focuses on exploring the nonlinear dispersion mechanism by considering the wave velocity factor to analyze the behavior of incompressible fluids. To do this, two analytical methodologies are suggested, and solutions are generated that simulate many varieties of traveling waves under particular constraining circumstances. The stability of the equation considered in the study has beenevaluated using a linear stability analysis method that examines the effects of small-amplitude perturbations on the solution. The nonlinear wave distribution in the *x* direction is examined by selecting one of these solutions, which is the model of the time-dependent behavior of incompressible fluids, and by assigning the proper values to the constants that keep the system stable. In addition, while comparing the advantages and limitations of the two analytical methods, the study also draws attention to the strong physical correlation between diffusion, absorption, and wave velocity. Since wave velocity is scientifically related to energy per unit space or time, it has been concluded that high-energy waves propagate and diffuse more rapidly. The steady-state stability of the equation, which is the model of the nonlinear distribution, is determined by stimulated Raman scattering, group velocity distribution, and eigen-phase modulation. It has been argued that fluid flow modeling, simulation, and use of analytical results can help throughout the design phase to produce safe and useful products.

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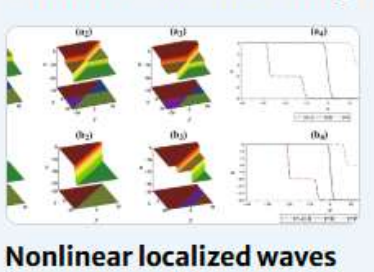

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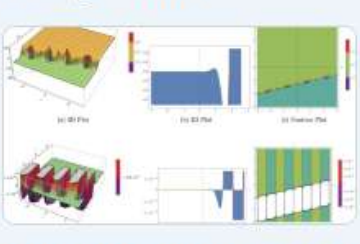

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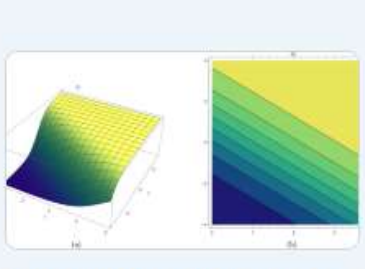

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Data Availability Statement

The authors confirm that the data underpinning the findings of this study are included within the article, with the figures providing a clear representation of the results.

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Ethics declarations

Conflict of interest

The authors declare that there are no financial, personal, or professional conflicts of interest that could have influenced the research, authorship, or publication of this article.

Ethical standards

The authors confirm that this work adheres to established ethical standards.

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Yokus, A., Isah, M.A. & Tuz, M. Analytical Investigation of Nonlinear Distribution Mechanisms in Fluid Dynamics through the Boiti-Leon-Manna-Pempinelli Equation. *Comput Math Model* (2025). https://doi.org/10.1007/s10598-025-09630-2

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Received	Revised	Accepted
10 July 2025	27 August 2025	29 August 2025

Published	Version of record
26 September 2025	26 September 2025

DOI
https://doi.org/10.1007/s10598-025-09630-2

Keywords

The modified \(\left(\frac{1}{G^{\prime}}\right)\)-expansion technique
The 3 + 1-dimensional Boiti–Leon–Manna–Pempinelli equation
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